

DSA 8301: Statistical Inference for Big Data

Lecture Notes

Topic 4: Asymptotic Properties of Estimators

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4 Asymptotic Properties of Estimators

Asymptotic theory studies the behavior of estimators as the sample size $n \rightarrow \infty$. It helps us analyze important properties such as consistency, limiting distributions, and efficiency. A key mathematical foundation for asymptotic analysis is the concept of **modes of convergence**.

4.1 Modes of Convergence

Modes of convergence describe different ways in which a sequence of random variables $\{X_n\}$ approaches a limiting random variable X .

4.1.1 Convergence in Probability

A sequence X_n converges in probability to X if:

$$P(|X_n - X| > \epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

for every $\epsilon > 0$.

Notation: $X_n \xrightarrow{p} X$

Intuition As n increases, the probability that X_n differs significantly from X becomes very small. In other words, X_n gets arbitrarily close to X with high probability.

Example 4.1.1

Deterministic Sequence Let $X_n = \frac{1}{n}$. Then:

$$P(|X_n - 0| > \epsilon) = P\left(\frac{1}{n} > \epsilon\right)$$

For sufficiently large n , $\frac{1}{n} < \epsilon$, hence:

$$P(|X_n - 0| > \epsilon) = 0$$

Thus:

$$X_n \xrightarrow{p} 0$$

Example 4.1.2

Example 2: Sample Mean Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Then:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Using Chebyshev's inequality:

$$P(|\bar{X}_n - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$$

Hence:

$$\bar{X}_n \xrightarrow{p} \mu$$

Importance Convergence in probability is central to the concept of **consistency of estimators**.

4.1.2 Almost Sure Convergence

A sequence of random variables $\{X_n\}$ converges almost surely (a.s.) to a random variable X if:

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

Notation:

$$X_n \xrightarrow{a.s.} X$$

Equivalent Definition $X_n \xrightarrow{a.s.} X$ if:

$$P\left(\left\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1$$

That is, for almost every outcome ω , the sequence $X_n(\omega)$ converges to $X(\omega)$.

Intuition Almost sure convergence means that, except possibly on a set of probability zero, the sequence behaves like an ordinary convergent sequence. In other words, for almost every sample path, X_n eventually gets arbitrarily close to X and stays close.

Example 4.1.3

Deterministic Sequence Let:

$$X_n = \frac{1}{n}$$

Then for every ω :

$$\lim_{n \rightarrow \infty} X_n = 0$$

Hence:

$$P\left(\lim_{n \rightarrow \infty} X_n = 0\right) = 1$$

So:

$$X_n \xrightarrow{a.s.} 0$$

Example 4.1.4

Strong Law of Large Numbers Let $X_1, X_2, \dots$ be i.i.d. random variables with:

$$E[X_i] = \mu$$

Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Then:

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

Sketch of Proof (Kolmogorov's Strong Law):

Let $S_n = \sum_{i=1}^n (X_i - \mu)$. Then:

$$\bar{X}_n - \mu = \frac{S_n}{n}$$

Using Kolmogorov's inequality and Borel-Cantelli Lemma, one can show:

$$\frac{S_n}{n} \rightarrow 0 \quad \text{almost surely}$$

Hence:

$$\bar{X}_n \rightarrow \mu \quad \text{a.s.}$$

4.1.3 Convergence in Distribution

A sequence of random variables $\{X_n\}$ is said to converge in distribution to a random variable X if:

$$F_{X_n}(x) \rightarrow F_X(x)$$

for all points x at which $F_X(x)$ is continuous.

Notation:

$$X_n \xrightarrow{d} X$$

Intuition Convergence in distribution means that the *entire distribution* of X_n approaches that of X as $n \rightarrow \infty$. This does not require X_n to get close to X for each outcome; instead, it only requires that probabilities of events behave similarly in the limit.

In simple terms:

- The shape of the distribution of X_n becomes closer to that of X .
- Individual sample paths may not converge.
- It is the weakest commonly used mode of convergence.

Equivalent Definitions: Using Expectations of Continuous Functions: $X_n \xrightarrow{d} X$ if for every bounded continuous function g :

$$E[g(X_n)] \rightarrow E[g(X)]$$

Example 4.1.5**Example 1: Degenerate Convergence** Let:

$$X_n \sim N(0, 1/n)$$

That is, X_n is normally distributed with mean 0 and variance $\frac{1}{n}$. As $n \rightarrow \infty$, the variance shrinks to 0, meaning the distribution becomes more and more concentrated around 0.

Step 1: Write the Distribution Function The cumulative distribution function (CDF) of X_n is:

$$F_{X_n}(x) = P(X_n \leq x)$$

Since $X_n \sim N(0, 1/n)$, we standardize:

$$Z_n = \frac{X_n}{\sqrt{1/n}} = \sqrt{n}X_n \sim N(0, 1)$$

Thus:

$$F_{X_n}(x) = P(X_n \leq x) = P(\sqrt{n}X_n \leq \sqrt{n}x) = P(Z_n \leq \sqrt{n}x)$$

Hence:

$$F_{X_n}(x) = \Phi(\sqrt{n}x)$$

where $\Phi(\cdot)$ is the standard normal CDF.

Step 2: Take the Limit We now evaluate:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = \lim_{n \rightarrow \infty} \Phi(\sqrt{n}x)$$

Consider three cases:

Case 1: $x < 0$

Then $\sqrt{n}x \rightarrow -\infty$ as $n \rightarrow \infty$. Since:

$$\Phi(t) \rightarrow 0 \quad \text{as } t \rightarrow -\infty$$

we have:

$$F_{X_n}(x) \rightarrow 0$$

Case 2: $x > 0$

Then $\sqrt{n}x \rightarrow +\infty$ as $n \rightarrow \infty$. Since:

$$\Phi(t) \rightarrow 1 \quad \text{as } t \rightarrow +\infty$$

we get:

$$F_{X_n}(x) \rightarrow 1$$

Case 3: $x = 0$

Then:

$$F_{X_n}(0) = \Phi(0) = \frac{1}{2}$$

So:

$$\lim_{n \rightarrow \infty} F_{X_n}(0) = \frac{1}{2}$$

Step 3: Identify the Limiting Distribution Combining the results:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}, & x = 0 \\ 1, & x > 0 \end{cases}$$

However, convergence in distribution only requires convergence at points where the limiting CDF is **continuous**.

Now consider the distribution:

$$X = 0 \quad (\text{constant random variable})$$

Its CDF is:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

This function has a jump (discontinuity) at $x = 0$.

Step 4: Justification of Convergence For all points where $F_X(x)$ is continuous (i.e., all $x \neq 0$):

$$F_{X_n}(x) \rightarrow F_X(x)$$

Thus, by definition of convergence in distribution:

$$X_n \xrightarrow{d} 0$$

Step 5: Intuition

- As n increases, the variance $\frac{1}{n}$ shrinks to 0.
- The normal distribution becomes more and more concentrated around its mean 0.
- In the limit, all probability mass collapses at a single point 0.
- The limiting distribution is therefore a **degenerate distribution** at 0.

Step 6: Additional Insight We can also observe that:

$$\text{Var}(X_n) = \frac{1}{n} \rightarrow 0$$

This suggests not only convergence in distribution, but also:

$$X_n \xrightarrow{p} 0$$

since shrinking variance implies concentration around the mean.

Conclusion

- $X_n \sim N(0, 1/n)$ becomes increasingly concentrated at 0.
- Its CDF converges pointwise (except at 0) to that of a constant random variable.
- Therefore:

$$X_n \xrightarrow{d} 0$$

Example 4.1.6

Example 2: Central Limit Theorem (Detailed) Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with:

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

We standardize $\bar{X}_n$ to obtain:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

Step 1: Mean and Variance of the Sample Mean First compute:

$$E[\bar{X}_n] = \mu$$

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Thus:

$\bar{X}_n$ becomes more concentrated around μ as $n \rightarrow \infty$

Step 2: Standardization To study the distribution, we standardize:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

This ensures:

$$E[Z_n] = 0, \quad \text{Var}(Z_n) = 1$$

Step 3: Apply the Central Limit Theorem The Central Limit Theorem states that:

$$Z_n \xrightarrow{d} N(0, 1)$$

That is:

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \Phi(z)$$

for all $z \in \mathbb{R}$, where $\Phi(z)$ is the standard normal CDF.

Step 4: Interpretation of the Result Even if the original variables X_i are:

- not normally distributed,
- skewed,
- or discrete,

the distribution of Z_n approaches the standard normal distribution as n becomes large.

Step 5: Equivalent Formulation The result can also be written as:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

This shows the rate of convergence: the deviation shrinks at order $1/\sqrt{n}$.

Step 6: Practical Approximation For large n , we approximate:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Thus:

$$P(\bar{X}_n \leq x) \approx \Phi\left(\frac{x - \mu}{\sigma/\sqrt{n}}\right)$$

Example (Concrete Case) Suppose:

$$X_i \sim \text{Bernoulli}(p)$$

Then:

$$\mu = p, \quad \sigma^2 = p(1 - p)$$

The sample mean $\bar{X}_n$ is the sample proportion. By CLT:

$$\frac{\bar{X}_n - p}{\sqrt{p(1 - p)/n}} \xrightarrow{d} N(0, 1)$$

Thus, for large n :

$$\bar{X}_n \approx N\left(p, \frac{p(1 - p)}{n}\right)$$

Key Insights

- The CLT explains why normal distributions appear frequently in statistics.
- It justifies using normal approximations for large samples.
- The rate of convergence is $\sqrt{n}$.
- It applies broadly, requiring only finite mean and variance.

Example 4.1.7

No Convergence in Probability Let $\{X_n\}$ be a sequence of random variables such that:

$$X_n \sim N(0, 1) \quad \text{for all } n$$

That is, each X_n has the same standard normal distribution.

Step 1: Convergence in Distribution Since the distribution of X_n does not depend on n , we have:

$$F_{X_n}(x) = \Phi(x) \quad \text{for all } n$$

Thus:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = \Phi(x)$$

which is the CDF of $N(0, 1)$. Therefore:

$$X_n \xrightarrow{d} N(0, 1)$$

Step 2: Does X_n Converge in Probability? We now investigate whether there exists a random variable X such that:

$$X_n \xrightarrow{p} X$$

Key Fact If $X_n \xrightarrow{p} X$, then it must also be true that:

$$X_n \xrightarrow{d} X$$

But we already know:

$$X_n \xrightarrow{d} N(0, 1)$$

Hence, if convergence in probability were to hold, the limiting random variable must be:

$$X \sim N(0, 1)$$

Step 3: Test Convergence in Probability To check convergence in probability, we examine:

$$P(|X_n - X| > \epsilon)$$

However, since X_n and X have the same distribution but are not necessarily equal, consider the case where X is independent of X_n .

Then:

$$X_n - X \sim N(0, 2)$$

Thus:

$$P(|X_n - X| > \epsilon) = P(|Z| > \epsilon/\sqrt{2})$$

where $Z \sim N(0, 1)$.

This probability is a fixed positive constant (independent of n), hence:

$$P(|X_n - X| > \epsilon) \not\rightarrow 0$$

Therefore:

$$X_n \not\xrightarrow{p} X$$

Step 4: Intuition

- The distribution of X_n does not change as n increases.
- There is no “shrinking” or concentration of probability mass.
- The random variables keep fluctuating with the same spread.
- Hence, they do not settle around any single random variable.

Step 5: Important Insight Convergence in distribution only requires:

- Matching of distributions in the limit,
- Not closeness of individual realizations.

Thus, even though:

$$X_n \xrightarrow{d} N(0, 1)$$

there is no guarantee of convergence in probability.

Conclusion

- $X_n \sim N(0, 1)$ for all n implies:

$$X_n \xrightarrow{d} N(0, 1)$$

- However, X_n does not converge in probability to any random variable.
- This example demonstrates that:

$$X_n \xrightarrow{d} X \not\Rightarrow X_n \xrightarrow{p} X$$

Example 4.1.8

Example 4: Convergence of Indicators (Detailed) Let $\{X_n\}$ be a sequence of random variables defined by:

$$X_n = \begin{cases} 1, & \text{with probability } \frac{1}{n} \\ 0, & \text{with probability } 1 - \frac{1}{n} \end{cases}$$

Step 1: Understand the Distribution Each X_n is a Bernoulli random variable with parameter:

$$p_n = \frac{1}{n}$$

Thus:

$$E[X_n] = \frac{1}{n}, \quad \text{Var}(X_n) = \frac{1}{n} \left(1 - \frac{1}{n}\right)$$

As $n \rightarrow \infty$:

$$E[X_n] \rightarrow 0, \quad \text{Var}(X_n) \rightarrow 0$$

This suggests that X_n becomes increasingly concentrated at 0.

Step 2: Compute the Distribution Function The cumulative distribution function (CDF) of X_n is:

$$F_{X_n}(x) = \begin{cases} 0, & x < 0 \\ P(X_n = 0) = 1 - \frac{1}{n}, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

Step 3: Take the Limit We now evaluate $\lim_{n \rightarrow \infty} F_{X_n}(x)$.

Case 1: $x < 0$

$$F_{X_n}(x) = 0 \quad \Rightarrow \quad \lim_{n \rightarrow \infty} F_{X_n}(x) = 0$$

Case 2: $0 \leq x < 1$

$$F_{X_n}(x) = 1 - \frac{1}{n} \rightarrow 1$$

Case 3: $x \geq 1$

$$F_{X_n}(x) = 1 \quad \Rightarrow \quad \lim_{n \rightarrow \infty} F_{X_n}(x) = 1$$

Thus, the limiting CDF is:

$$F(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Step 4: Identify the Limiting Distribution The limiting function $F(x)$ is the CDF of a constant random variable:

$$X = 0$$

Thus:

$$X_n \xrightarrow{d} 0$$

Step 5: Convergence in Probability We now check convergence in probability:

$$P(|X_n - 0| > \epsilon)$$

For $0 < \epsilon \leq 1$:

$$P(|X_n| > \epsilon) = P(X_n = 1) = \frac{1}{n} \rightarrow 0$$

Thus:

$$X_n \xrightarrow{p} 0$$

Step 6: Convergence Almost Surely Consider the events:

$$A_n = \{X_n = 1\}$$

Then:

$$P(A_n) = \frac{1}{n}$$

Since:

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

and the events are independent, by the second Borel–Cantelli lemma:

$$P(X_n = 1 \text{ infinitely often}) = 1$$

Thus, X_n does **not** converge almost surely to 0.

Step 7: Intuition

- The probability of $X_n = 1$ becomes smaller and smaller.
- Most of the time, $X_n = 0$ for large n .
- However, the event $X_n = 1$ still occurs infinitely often (though rarely).
- Hence:
 - The distribution collapses to 0 (convergence in distribution),
 - The probability of deviation goes to zero (convergence in probability),
 - But exact pathwise convergence fails (no almost sure convergence).

Conclusion

- $X_n \xrightarrow{d} 0$
- $X_n \xrightarrow{p} 0$
- $X_n \not\xrightarrow{a.s.} 0$

This example clearly distinguishes the three modes of convergence.

4.1.4 Relationship Between Modes of Convergence

In probability theory and asymptotic statistics, the different modes of convergence—almost sure, in probability, and in distribution—are related in a hierarchical way. The standard hierarchy is:

$$X_n \xrightarrow{a.s.} X \quad \Rightarrow \quad X_n \xrightarrow{p} X \quad \Rightarrow \quad X_n \xrightarrow{d} X$$

We now explain each step carefully.

1. Almost Sure Convergence Implies Convergence in Probability **Definition:**

$X_n \xrightarrow{a.s.} X$ if

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

Implication: Almost sure convergence is the strongest form of convergence because it requires that *for almost every individual outcome*, $X_n(\omega)$ eventually equals $X(\omega)$.

Theoretical Argument:

For any $\epsilon > 0$, define the event:

$$A_n^\epsilon = \{|X_n - X| > \epsilon\}$$

Almost sure convergence implies that for almost every outcome, $|X_n - X| \leq \epsilon$ eventually. Hence:

$$P(A_n^\epsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

But this is exactly the definition of convergence in probability:

$$X_n \xrightarrow{p} X$$

Intuition: If every sample path eventually reaches X , then the probability of deviating from X by more than ϵ goes to zero.

2. Convergence in Probability Implies Convergence in Distribution

Definition: $X_n \xrightarrow{p} X$ if

$$P(|X_n - X| > \epsilon) \rightarrow 0 \quad \forall \epsilon > 0$$

Implication: Convergence in probability ensures that X_n gets arbitrarily close to X with high probability as n grows.

Theoretical Argument:

For any continuity point x of F_X , the cumulative distribution function of X :

$$F_{X_n}(x) = P(X_n \leq x)$$

Since X_n is close to X with high probability, we have:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

Hence, convergence in probability implies convergence in distribution:

$$X_n \xrightarrow{d} X$$

Intuition: If the values of X_n cluster around X , then the distribution of X_n approaches the distribution of X .

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4.2 Law of Large Numbers and Central Limit Theorem

Asymptotic theory studies the behavior of sequences of random variables as the sample size $n \rightarrow \infty$. Two foundational results in this theory are the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT).

4.2.1 Law of Large Numbers (LLN)

The Law of Large Numbers describes how the sample average of independent and identically distributed (i.i.d.) random variables converges to the population mean as the sample size grows.

Weak Law of Large Numbers (WLLN)

Statement: Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean $\mu = E[X_i]$. Then the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges in probability to μ :

$$\bar{X}_n \xrightarrow{p} \mu$$

Proof Sketch (Using Chebyshev's Inequality):

- Let $\sigma^2 = \text{Var}(X_i) < \infty$.
- Compute the variance of the sample mean:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{\sigma^2}{n}$$

- Apply Chebyshev's inequality for any $\epsilon > 0$:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Hence, $\bar{X}_n \xrightarrow{p} \mu$.

Intuition: As the number of observations increases, the sample mean becomes increasingly concentrated around the true mean μ . Fluctuations average out.

Example 4.2.1

Consider a sequence of independent tosses of a fair coin. Let X_i denote the outcome of the i th toss:

$$X_i = \begin{cases} 1, & \text{if the } i\text{th toss is Heads} \\ 0, & \text{if the } i\text{th toss is Tails} \end{cases}$$

Step 1: Identify the Mean and Variance For a fair coin:

$$P(X_i = 1) = P(X_i = 0) = 0.5$$

The mean and variance of each X_i are:

$$\mu = E[X_i] = 1 \cdot 0.5 + 0 \cdot 0.5 = 0.5$$

$$\sigma^2 = \text{Var}(X_i) = 0.5 \cdot (1 - 0.5)^2 + 0.5 \cdot (0 - 0.5)^2 = 0.25$$

Step 2: Define the Sample Mean The sample mean of n tosses is:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Step 3: Apply the Weak Law of Large Numbers The Weak Law of Large Numbers (WLLN) states that:

$$\bar{X}_n \xrightarrow{p} \mu$$

Using Chebyshev's inequality for $\epsilon > 0$:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2/n}{\epsilon^2} = \frac{0.25}{n\epsilon^2}$$

As $n \rightarrow \infty$:

$$P(|\bar{X}_n - 0.5| \geq \epsilon) \rightarrow 0$$

This means the probability that the proportion of heads deviates from 0.5 by more than any fixed ϵ becomes negligible for large n .

Step 4: Numerical Illustration Suppose we toss the coin $n = 100$ times and want the probability that the sample proportion of heads deviates from 0.5 by more than $\epsilon = 0.1$:

$$P(|\bar{X}_{100} - 0.5| \geq 0.1) \leq \frac{0.25}{100 \cdot 0.1^2} = \frac{0.25}{1} = 0.25$$

For $n = 1000$:

$$P(|\bar{X}_{1000} - 0.5| \geq 0.1) \leq \frac{0.25}{1000 \cdot 0.1^2} = 0.025$$

Observation: As n increases, the probability of a significant deviation from 0.5 decreases rapidly.

Step 5: Intuition

- Each coin toss is random, but on average, heads occur half of the time.
- As we increase the number of tosses, the sample mean $\bar{X}_n$ becomes more concentrated around 0.5.
- For very large n , the proportion of heads is extremely likely to be very close to 0.5.

Step 6: Conclusion The WLLN guarantees that for a fair coin:

$$\bar{X}_n \xrightarrow{p} 0.5$$

Even though individual tosses are random, the average outcome stabilizes around the true mean as the number of tosses grows.

Strong Law of Large Numbers (SLLN) – Detailed

Statement: Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with finite mean $\mu = E[X_i]$. Then the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges **almost surely** to μ :

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

1. What Almost Sure Convergence Means in SLLN The notation $\bar{X}_n \xrightarrow{a.s.} \mu$ means:

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

- This is stronger than convergence in probability (Weak Law of Large Numbers).
- It states that for **almost every possible outcome** (sample path), the sequence of sample means $\bar{X}_n$ eventually gets arbitrarily close to μ and stays close as $n \rightarrow \infty$.

2. Intuition Behind SLLN

- Each individual observation X_i can fluctuate randomly.
- As we average more observations, random fluctuations "cancel out" along almost every sample path.
- Eventually, the sample mean stabilizes at the true mean μ and does not deviate significantly.
- The convergence is pathwise; it applies to each sequence of outcomes except for a set of probability zero.

Example 4.2.2

Worked Example: SLLN for Dice Rolls Problem:

Consider rolling a fair six-sided die repeatedly. Let X_i denote the outcome of the i th roll:

$$X_i \in \{1, 2, 3, 4, 5, 6\}, \quad P(X_i = k) = \frac{1}{6}, \quad k = 1, \dots, 6$$

Step 1: Compute the mean and variance of X_i

$$\mu = E[X_i] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

$$\sigma^2 = \text{Var}(X_i) = E[X_i^2] - (E[X_i])^2$$

$$E[X_i^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6} \approx 15.1667$$

$$\sigma^2 = 15.1667 - 3.5^2 = 15.1667 - 12.25 = 2.9167$$

Step 2: Define the sample mean

After n rolls, the sample mean is:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Step 3: Apply SLLN

By the Strong Law of Large Numbers:

$$\bar{X}_n \xrightarrow{a.s.} \mu = 3.5$$

Interpretation: For almost every possible infinite sequence of die rolls, the cumulative average of the outcomes will converge to 3.5 as $n \rightarrow \infty$.

Step 4: Numerical Illustration

Suppose we simulate the cumulative average for $n = 10, 50, 100, 500$ rolls:

Number of Rolls (n)	Cumulative Average ($\bar{X}_n$)
10	3.8
50	3.56
100	3.49
500	3.503

Observation: As the number of rolls increases, the cumulative average approaches 3.5.

Step 5: Intuition

- Initially, random fluctuations cause the average to deviate from 3.5.
- With more rolls, these fluctuations cancel out.
- Eventually, the average stabilizes very close to the true mean.
- This demonstrates pathwise convergence – along almost every sequence of die rolls, the sample mean converges to 3.5.

Conclusion:

The SLLN guarantees that no matter how the outcomes fluctuate initially, the long-term average of a fair die roll converges almost surely to its expected value 3.5.

4.2.2 Central Limit Theorem (CLT)

The CLT describes the distributional behavior of the standardized sample mean as $n \rightarrow \infty$.

Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean $\mu = E[X_i]$ and variance $\sigma^2 = \text{Var}(X_i) < \infty$. Then, the standardized sample mean:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

converges in distribution to a standard normal:

$$Z_n \xrightarrow{d} N(0, 1)$$

Interpretation:

- Regardless of the original distribution of X_i , the standardized sample mean becomes approximately normal for large n .
- The CLT explains why the normal distribution is ubiquitous in statistics.

Step 1: Mean and Variance of Z_n

$$E[Z_n] = 0, \quad \text{Var}(Z_n) = 1$$

Step 2: Approximation for Large n For large n :

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Example 1 (Dice Rolls): Let X_i be the outcome of the i th roll of a fair 6-sided die:

$$E[X_i] = 3.5, \quad \text{Var}(X_i) = \frac{35}{12}$$

For $n = 100$ rolls, the sample mean $\bar{X}_{100}$ is approximately:

$$\bar{X}_{100} \sim N\left(3.5, \frac{35}{12 \cdot 100}\right)$$

Hence, the probability that the average of 100 rolls is between 3 and 4 can be approximated using the normal distribution.

Example 2 (Bernoulli Trials): Let $X_i \sim \text{Bernoulli}(p)$. Then:

$$Z_n = \frac{\bar{X}_n - p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0, 1)$$

This justifies using normal approximations for sample proportions in surveys.

Step 3: Intuition

- LLN tells us the sample mean converges to the true mean.
- CLT refines this by describing how the sample mean fluctuates around the mean for large n .
- These fluctuations are normally distributed, regardless of the original X_i distribution.

4.3 Asymptotic Normality of Estimators

Definition:

An estimator $\hat{\theta}_n$ of a parameter θ is said to be **asymptotically normal** if, as the sample size $n \rightarrow \infty$:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, V)$$

where V is the **asymptotic variance** of $\hat{\theta}_n$.

—

1. Intuition

- For large n , the distribution of the estimator $\hat{\theta}_n$ becomes approximately normal, regardless of the original distribution of the data.
 - The factor $\sqrt{n}$ rescales the deviations of $\hat{\theta}_n$ from the true parameter θ .
 - As n increases, $\hat{\theta}_n$ concentrates around θ , and the scaled difference $\sqrt{n}(\hat{\theta}_n - \theta)$ converges in distribution to a normal distribution.
 - This result allows statisticians to construct approximate confidence intervals and hypothesis tests for large samples.
-

2. Relation to the Central Limit Theorem (CLT)

- The CLT provides the underlying mechanism for asymptotic normality.
- If $\hat{\theta}_n$ is the sample mean $\bar{X}_n$ of i.i.d. observations with mean θ and variance σ^2 , then by the CLT:

$$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$$

- Hence, the sample mean is a classic example of an asymptotically normal estimator.
-

3. General Form For a general estimator $\hat{\theta}_n$:

$$\hat{\theta}_n \approx N\left(\theta, \frac{V}{n}\right) \quad \text{for large } n$$

- The unscaled estimator $\hat{\theta}_n$ has variance shrinking at the rate $1/n$.
 - Scaling by $\sqrt{n}$ produces a limiting distribution with finite variance V .
-

Example 4.3.1**Example 1: Sample Mean – Detailed Setup:**

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with:

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

The sample mean is defined as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

We want to study the distribution of $\bar{X}_n$ for large n .

—
Step 1: Standardize the sample mean To analyze its limiting behavior, consider the scaled difference:

$$\sqrt{n}(\bar{X}_n - \mu)$$

- This scaling compensates for the fact that $\bar{X}_n$ becomes closer to μ as n grows.
- Without scaling, $\bar{X}_n - \mu \rightarrow 0$ by the Law of Large Numbers, giving no information about the variability.

—
Step 2: Apply the Central Limit Theorem (CLT) The CLT states:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

- This is an instance of **asymptotic normality**.
- The sample mean is asymptotically normal with mean μ and variance σ^2/n .

—
Step 3: Interpret the result For large n , the sample mean $\bar{X}_n$ is approximately:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

- The variance decreases as $1/n$, meaning the estimator becomes more precise with larger sample size.
 - The distribution is normal regardless of the original distribution of X_i (provided finite variance).
-

Step 4: Numerical Illustration Suppose X_i has mean $\mu = 10$ and variance $\sigma^2 = 9$.

- For $n = 100$ observations:

$$\bar{X}_{100} \approx N\left(10, \frac{9}{100}\right) = N(10, 0.09)$$

- Standard deviation of $\bar{X}_{100}$: $\sqrt{0.09} = 0.3$

Confidence Interval Example:

A 95% confidence interval for μ based on $\bar{X}_{100}$:

$$\bar{X}_{100} \pm 1.96 \cdot 0.3 = \bar{X}_{100} \pm 0.588$$

- This means that with large n , we can be 95% confident that μ lies within 0.588 units of the sample mean.

Step 5: Intuition

- As n grows, the sample mean “fluctuates less” around μ .
- The fluctuations are approximately normal when rescaled by $\sqrt{n}$.
- This is the fundamental reason why the normal distribution appears in many statistical inference procedures.

Step 6: Summary

- The sample mean is an asymptotically normal estimator.
- Scaling by $\sqrt{n}$ gives a limiting distribution $N(0, \sigma^2)$.
- For large n , we can approximate the distribution of $\bar{X}_n$ by $N(\mu, \sigma^2/n)$.
- This allows construction of confidence intervals and hypothesis tests even if the original data is not normal.

Example 4.3.2

Sample Proportion – Detailed Setup:

Let $X_1, X_2, \dots, X_n$ be independent Bernoulli random variables with success probability p :

$$X_i = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1 - p \end{cases}$$

The sample proportion of successes is:

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$$

—

Step 1: Compute mean and variance

- Mean: $E[X_i] = p$
- Variance: $\text{Var}(X_i) = p(1 - p)$

Hence, $\hat{p}_n$ is an estimator of p .

—

Step 2: Apply the Central Limit Theorem The CLT implies that for large n , the sample mean of i.i.d. variables is asymptotically normal:

$$\sqrt{n}(\hat{p}_n - p) = \sqrt{n}(\bar{X}_n - p) \xrightarrow{d} N(0, p(1 - p))$$

—

Step 3: Interpretation

- The sample proportion is approximately normal for large n .
- The asymptotic variance of $\hat{p}_n$ decreases as $1/n$:

$$\text{Var}(\hat{p}_n) \approx \frac{p(1 - p)}{n}$$

- This allows us to construct approximate confidence intervals for p :

$$\hat{p}_n \pm z_{\alpha/2} \sqrt{\frac{p(1 - p)}{n}}$$

where $z_{\alpha/2}$ is the critical value from the standard normal distribution.

—

Step 4: Numerical Illustration Suppose a coin is tossed $n = 100$ times, and we let $p = 0.5$ (fair coin).

- The sample proportion $\hat{p}_{100}$ is approximately normal:

$$\hat{p}_{100} \sim N\left(0.5, \frac{0.5 \cdot 0.5}{100}\right) = N(0.5, 0.0025)$$

- Standard deviation: $\sqrt{0.0025} = 0.05$
- 95% confidence interval:

$$\hat{p}_{100} \pm 1.96 \cdot 0.05 = \hat{p}_{100} \pm 0.098$$

Observation: As n increases, the sample proportion becomes more concentrated around the true proportion p .

—

Step 5: Key Points

- Sample proportions are asymptotically normal for large n .
- The rate of convergence depends on $1/\sqrt{n}$.
- This approximation is widely used in surveys, polling, and binomial inference.
- Even though the original distribution is discrete (Bernoulli), the asymptotic distribution of $\hat{p}_n$ is continuous (normal).

Example 4.3.3

Worked Example: Asymptotic Normality of the Sample Mean Problem:

Suppose $X_1, X_2, \dots, X_n$ are independent and identically distributed (i.i.d.) with:

$$E[X_i] = \theta = 5, \quad \text{Var}(X_i) = \sigma^2 = 4$$

Let the sample mean be the estimator:

$$\hat{\theta}_n = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

We want to describe the asymptotic distribution of $\hat{\theta}_n$ and use it for inference.

Step 1: Compute the scaled difference To study the asymptotic behavior, consider the scaled estimator:

$$\sqrt{n}(\hat{\theta}_n - \theta) = \sqrt{n}(\bar{X}_n - 5)$$

- Scaling by $\sqrt{n}$ ensures a non-degenerate limiting distribution as $n \rightarrow \infty$.

Step 2: Identify the asymptotic variance

- For the sample mean, the asymptotic variance is equal to the population variance:

$$V = \sigma^2 = 4$$

- This variance corresponds to the limiting normal distribution of the scaled estimator.

Step 3: Apply asymptotic normality By the Central Limit Theorem (CLT) or general asymptotic normality results:

$$\sqrt{n}(\hat{\theta}_n - 5) \xrightarrow{d} N(0, 4)$$

- Interpretation: For large n , the distribution of the scaled deviation of $\hat{\theta}_n$ from θ is approximately normal with mean 0 and variance 4.

Step 4: Approximate the distribution of the unscaled sample mean For large n , the unscaled sample mean is approximately:

$$\hat{\theta}_n \sim N\left(\theta, \frac{V}{n}\right) = N\left(5, \frac{4}{n}\right)$$

- Example: If $n = 100$,

$$\hat{\theta}_{100} \sim N\left(5, \frac{4}{100}\right) = N(5, 0.04)$$

- Standard deviation: $\sqrt{0.04} = 0.2$

Step 5: Construct confidence intervals A 95% confidence interval for θ is:

$$\hat{\theta}_{100} \pm z_{0.025} \cdot \sqrt{\frac{4}{100}} = \hat{\theta}_{100} \pm 1.96 \cdot 0.2 = \hat{\theta}_{100} \pm 0.392$$

- Interpretation: With large sample size, the true mean $\theta = 5$ is likely to lie within 0.392 units of the observed sample mean.

Step 6: Step-by-step intuition

1. The sample mean $\hat{\theta}_n$ gets closer to the true mean θ as n increases (Law of Large Numbers).
2. Scaling by $\sqrt{n}$ produces a limiting normal distribution with finite variance.
3. The asymptotic normality allows approximation of probabilities, confidence intervals, and hypothesis tests for large n .
4. As n increases, the standard error $\sqrt{V/n}$ decreases, making $\hat{\theta}_n$ more precise.

Step 7: Summary

- $\hat{\theta}_n$ is asymptotically normal: $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$.
- For large samples, we can approximate $\hat{\theta}_n \sim N(\theta, \sigma^2/n)$.
- This example illustrates how asymptotic theory facilitates practical statistical inference.

Try Your Knowledge 4.1

5 6. Past Exam Questions

1. Define and explain the following modes of convergence of random variables. Provide the formal definition, notation, and a brief intuition for each case.
 - (a) Convergence in probability
 - (b) Almost sure convergence
 - (c) Convergence in distribution

2. State the Weak Law of Large Numbers (WLLN) formally. Then, provide a proof of the WLLN using either Chebyshev's inequality or another standard approach. Explain the significance of this result in the context of estimating population means from sample data.

3. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables with mean μ and variance σ^2 . Show that the standardized sample mean converges in distribution to the standard normal distribution:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Explain the steps clearly and interpret the result in the context of large-sample inference.

4. Prove that the sample mean $\bar{X}_n$ is a consistent estimator of the population mean μ . Include the definitions used, the method of proof, and an intuitive explanation of why the estimator becomes increasingly accurate as the sample size increases.
5. Explain the concept of asymptotic normality of an estimator $\hat{\theta}_n$. Define the asymptotic distribution and scaling. Give a concrete example, such as the sample mean or sample proportion, showing how $\hat{\theta}_n$ is approximately normal for large n . Include interpretation of the result.
6. Define asymptotic efficiency of an estimator. Explain why it is important in choosing between competing estimators. Provide an example or scenario where one estimator is asymptotically more efficient than another, and discuss the practical implications for large-sample inference.
7. Suppose an estimator $\hat{\theta}_n$ satisfies:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, 4)$$

Determine the approximate distribution of $\hat{\theta}_n$ for a large sample size n . Show the derivation step by step and explain how this result can be used to construct confidence intervals for θ .

8. Using Chebyshev's inequality, prove the Weak Law of Large Numbers. Include all steps clearly, and explain why Chebyshev's inequality provides a bound that leads to convergence in probability. Also, discuss the difference between the weak and strong laws in terms of convergence.